

Protocol 1

Outline:

- Step 1.** Identification of the given target protein sequence by using **BLAST**
- Step 2.** Retrieving information about the target protein sequence and its family through **UniProt**
- Step 3.** Searching the template protein of the target sequence using **BLAST**
- Step 4.** Sequence alignment of the target and the template sequence through **TCoffee server**
- Step 5.** Construction of homology model of target protein using **Modeller**
- Step 6.** Evaluation of the built model by **PROCHECK**, **ERRAT**, **Ramachanderan plot**

Assignment Deadline: Feb 21, 2025

Overview:

The protocol will start with an unknown protein sequence. You have to identify, model and finally evaluate the target protein.

Step 1- Identification of target sequence

(5/100 points)

Start with identifying this protein.

```
>UNKNOWN
METPLNSQKQLSACEDGDDCEGV LQKV VPTPGDKVEGQISNGYSAIPSGAGDRHSIPTTTTLVAELHQGER
ETWGKKVDFLLSVIGYAVDLGNVWRFPYICYQNGGGAFLPYTIMAIFGGIPLFYMEALGQYHRNGCISIW
RKICPIFKGIGYAIICIAFYIASYYNTIMAWALYYLISFTDQLPWTSCKNSWNTGNCTNYFSEDNIT
WTLHSTSPAEEFYTRHVLQIHRSKGLQDLGGISWQLALCIMLIFTVIYFSIWKGVKTSQKVVWVTATFPY
IILSVLLVRGATLPGAWRGVLFYLPKNWQKLLETGVWIDAAAQIFSLGPGFGVLLAFASYNKFNNCYQDA
LVTSVVNCMTSFVSGFVIFTVLGYMAEMRNEDVSEVAKDAGPSLLFITYAEAIAANMPASTFFAIIFFLML
ITLGLDSTFAGLEGVITAVLDEFPHVWAKRRERFVLAVVITCFFGSLVTLTFGGAYVVKLLEEYAPAVLT
VALIEAVAVSWFYGITQFCRDVKEMLGFS PGWFWRICWVAISPLFLLFIICSFLMSPPQLRLFQYNYPYW
SIILGYCIGTSSFICIPTYIAYRLIITPGTFKEIISITPETPTEIPCGDIRLNAV
```

- A. Use BLASTp server, run it for 1 iteration and use the non-redundant database as the search space (<https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins>).
- B. Identify the target protein. Provide name of the protein and also provide name of the protein family.

Step 2- Information about the target sequence (points)**(15/100**

- Go to the UniProtKB (<https://www.uniprot.org/>) to retrieve the information about the target sequence.
- Provide the accession number of the protein to which the target sequence belongs.
- What is this protein family in general and target protein in particular?

Step 3- Template Search**(20/100 points)**

- In order to model the structure of this protein, we need to identify correct templates, so start a new PSI-BLAST search for 3 iterations or start a new search with only PDB database as the search set. Run the BLAST. Click the related structures link at the top half of the results page and select your templates based on sequence identity, aligned length and bit score. Provide result of the selected template.

The screenshot shows the NCBI BLAST search interface. The 'Enter accession number(s), g(x), or FASTA sequence(s)' field contains a FASTA sequence. The 'Job Title' is 'Unknown'. The 'Choose Search Set' section is expanded, showing the 'Database' dropdown set to 'Protein Data Bank proteins (pdb)'. The 'Organism' field is empty. The 'Exclude' section has checkboxes for 'Models (MM/XP)', 'Non-redundant RefSeq proteins (WPI)', and 'Uncultured/environmental sample sequences'. The 'Program Selection' section shows 'blastp (protein-protein BLAST)' selected, with other options like 'PSI-BLAST (Position-Specific Iterated BLAST)', 'PHI-BLAST (Pattern Hs Initiated BLAST)', and 'DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)'.

OR

Enter accession number(s), gis(s), or FASTA sequence(s)

FASTA sequence: `Q9H0L3:Q9H0L3_P1[1-100]...Q9H0L3_P1[101-200]...Q9H0L3_P1[201-300]...Q9H0L3_P1[301-400]...Q9H0L3_P1[401-500]...Q9H0L3_P1[501-600]...Q9H0L3_P1[601-700]...Q9H0L3_P1[701-800]...Q9H0L3_P1[801-900]...Q9H0L3_P1[901-1000]`

Database: Non-redundant protein sequences (nr)

Organism: Enter organism name or id - completion will be suggested

Exclude: ☒ Models (MMXP) ☐ Non-redundant RefSeq proteins (RSP) ☐ Unstructured/environmental sample sequences

Program Selection: ☒ Quick BLASTP (Accelerated protein-protein BLAST) ☐ blastp (protein-protein BLAST) ☐ PSI-BLAST (Position-Specific Iterated BLAST) ☐ PHI-BLAST (Pattern-Hit Initiated BLAST) ☐ DELTA-BLAST (Domain-Enhanced Lookup Time Accelerated BLAST)

- Take a screenshot of the BLAST results and add a screenshot to this file.
- Select your templates (in order to keep things simpler, select only one template).
- Why did you select this template and not the others, you should be able to justify your choices by logical arguments.
- Download the PDB file of template (<https://www.rcsb.org/>) and save the respective sequence in FASTA format.

Step 4- Multiple Sequence Alignment

(20/100 points)

- Generate sequence alignments between template and target sequence by using an alignment program like TCOFFEE (<http://tcoffee.crg.cat/>). Select the regular 3DCoffee option to do the structural alignment (Espresso).
- Why did we do a structural alignment instead of a regular alignment?
- Paste the target and template sequences in FASTA format in the sequences input search box. Use the default input settings. The output options will include fasta_aln and clustalw_aln.
- Highlight the conserved regions, identify the gaps and paste a snapshot of the aligned sequences in your assignment.

Step 5- Homology Modeling

(20/100 points)

Note: Modeller is case-sensitive so always use consistent capitalization when referencing files and sequences. Stick to lowercase for PDB codes and uppercase for chain IDs. Double-check alignment files to ensure identifiers match exactly. Ensure this to avoid errors.

- Download the alignment result in FASTA format and convert it into PIR format. You can take help from <https://salilab.org/modeller/tutorial/basic.html> (step 1 of tutorial) for generating the PIR format of the aligned sequences.
PIR format should look like this:

```
>P1;TvLDH
sequence:TvLDH:::::0.00: 0.00
MSEAAHVLITGAAGQIGYILSHWIASGELYGDRQVYLHLLDIPPAMNRLTALTMELEDCAFPHLAGFVATTDPKA
AFKIDIDCAFLVASMPLKPGQVRADLISSNSVIFKNTGEYLSKWAKPSVKVLVIGNPDNTNCEIAMLHARNLKPEN
FSSLSMLDQNRAYYEVASKLGVDVKDVHDIIVWGNHGESMVADLTQATFTKEGKTQKVVDVLDHDYVFDTFKKI
GHRAWDILEHRGFTSAASPTKAAIQHMKAWLFGTAPGEVLSMGIPVPEGNPYGIKPGVWFSPFCNVDKEGKIHVV
EGFKVNDWLREKLDLFEKDLFHEKEIALNHLAQGG*
```

- B. Take a snapshot and paste in the file. What information does the header of PIR format provides?
- C. After converting the file into PIR format change the file extension to ali and save the target sequence as target.ali
- D. Now go to the third step of the tutorial (<https://salilab.org/modeller/tutorial/basic.html>) where you'll find the code snippet with the title "*align2d.py*" then create a python file (align.py) and paste this code. Open modeler and using the command cd change the directory to where "*align2d.py*" and ali files are saved (make sure they are in the same directory). Then type align2d.py and press enter. This will automatically generate another ali file with the information about the first and last residues so you can skip the step and jump to step F
- E. Convert the PIR file into ALI format to be used as an input to Modeller and save it as "alignment_file.ali". Details of the ALI file format can be found here (<https://salilab.org/modeller/9v2/manual/node412.html>). Sometimes the sequence of a protein and the structure sequence are not similar as some residues have been mutated or are missing from the structure for various reasons. Modeller however will not like this and usually complain that residues between structure and sequence are not same. So in order to complete our conversion from PIR to ALI format correctly we need to look at the template PDB file. Use Swiss-PdbViewer for this.
- F. Open pdb file in SPDBViewer, and look at the starting and ending residue numbers in control panel, and enter this information into the ALI file in the appropriate field.
- G. Select the chain A residues in the control panel and go to
File -> Save -> Sequence (fasta format)
 to save the structure sequence of your template.
 In order to see if there are any differences in both sequences, align them using any alignment tool e.g. <http://www.ebi.ac.uk/Tools/emboss/align/index.html>. Paste the final alignment file here as a screenshot.
- H. Create a folder. Keep the PDB file of template structure, alignment_file.ali and fasta sequence of target sequence in this folder. Create a python script according to your input files (this will be the model-single.py file from the Modeller tutorial). You will however need to change the names for the input files in this python script.
- I. Run Modeller using model-single.py script wait for your five models to be created
 1. Open a command line prompt. Click on the 'Modeller' link on your Start Menu.
 2. This will give you a Windows Command Prompt, set up for you to run Modeller.

3. Change to the directory containing the script and alignment files you created earlier, using the 'cd' command.
4. Run MODELLER itself by typing the following at the command prompt:
Mod9.21 model-single.py

Step 6- Model Evaluation**(20/100 points)**

- A. Evaluate the model through PROCHECK and ERRAT scoring function (<http://servicesn.mbi.ucla.edu/SAVES/>).
 1. Provide the ramachandran plots statistics (i.e., residues in the favorable, generously allowed and not-allowed regions). Are there any residues in the outlier region?
 2. Provide the ERRAT score value for the target sequence.
 3. Looking at the 3D structure of the structure, identify the number of helices and beta sheets.
 4. Describe the normal physiological role of the identified protein along with its malfunctioning in disease condition.